GRAVITYSORT · TRACEABILITY MATRIX

GravitySort traceability matrix

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/gravitysort		
Source	docs/03b-traceability.md @ d6d285d		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.2	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.1	2026-10-25	Amish Chadha	Generated by .kit/trace.py

GravitySort traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 14 user requirements, 16 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001	PR-001	No mercury	GVS-CAL-001	Design review of the flowsheet, TRL 3	Met (design review)
UR-002	PR-002	Throughput	GVS-CAL-001	Analysis, bowl capacity and cycle (GVS-CAL-001 section 3), TRL 3; timed run at TRL 4	Met on paper: 200 kg/h; 1.55 t per shift
UR-003, UR-014	PR-003	Bowl G and speed shown	GVS-CAL-001, GVS-BLD-001	Analysis, drive ratio (GVS-CAL-001 section 2), TRL 3; inspection of the display at TRL 4	Met on paper: 40 to 81 G at a 50 to 71 rpm cadence; speed display (BOM line 19)
UR-003	PR-004	Fine-gold recovery	GVS-CAL-001, GVS-BLD-001, GVS-STM-001	Test with characterized or spiked ore, TRL 4	Not verifiable at TRL 3; at risk.** Settling check passes (STM S2); bed retention not calculable (STM S3)
UR-004	PR-005	Smeltable concentrate	GVS-CAL-001, GVS-BLD-001, GVS-STM-001	Analysis, mass balance (GVS-CAL-001 section 8), TRL 3; test at TRL 4	At risk:** pull 0.33 % met; 5.3 kg needs 53:1 on the table, likely two passes (STM S15)
UR-005	PR-006	Pedal power	GVS-CAL-001, GVS-BLD-001, GVS-STM-001	Analysis, power estimate (GVS-CAL-001 section 7), TRL 3; measured at TRL 4	At risk:** 48.2 W at 60 G met; 62.8 W at 80 G not; union seal drag assumed
UR-006	PR-007	Motor option	GVS-CAL-001	Analysis (GVS-CAL-001 section 7), TRL 3	Met on paper: 5.2 times at 60 G, 4.0 times at 80 G; 0.55 kWh per shift
UR-007	PR-008	Water use and quality	GVS-CAL-001, GVS-BLD-001, GVS-STM-001	Analysis, water balance and supply (GVS-CAL-001 sections 4 and 6), TRL 3	Met on paper: 1.19 m3/h; 120 mesh screen filter on the fluidization hose (GVS-DEC-001 F2); with the 3/4 in valve and rotameter (F6), 12.6 kPa at the union and 10.0 kPa net at the lowest ring at 600 rpm at mid-tank
UR-008	PR-009	Parts cost	GVS-CAL-001	Analysis, priced BOM (GVS-CAL-001 section 12), TRL 3	Estimated cost $594: **over the value-engineering target by USD 139
UR-009	PR-010	Local workshop build	GVS-CAL-001, GVS-STM-001	Design review of every part (GVS-BLD-001), TRL 3; build at TRL 4	Met (design review); casting route unproven (STM S25)

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-010	PR-011	Transport mass	GVS-CAL-001, GVS-BLD-001, GVS-STM-001, GVS-OPT-001	Analysis, mass estimate (GVS-CAL-001 section 11), TRL 3; weighed at TRL 4	At risk:** 99.6 kg in six loads, heaviest 27.3 kg, only 0.4 kg of margin on an estimate (independent review); assembly time not verified
UR-011	PR-012	Guards, speed and stopping	GVS-CAL-001, GVS-BLD-001, GVS-STM-001	Design review, burst calculation (GVS-CAL-001 section 10), TRL 3	Met on paper: guards, lid interlock and disc brake (0.6 s stop); motor capped; lowest safety factor 16 at 1,200 rpm
UR-012	PR-013	Quick, secure clean-up	GVS-CAL-001, GVS-BLD-001	Design review, TRL 3; timed trial at TRL 4	Met on paper: toolless lid clamps; flush container and concentrate box with padlock hasps (GVS-DEC-001); time not verified
UR-013	PR-014	Liner life and replacement	GVS-CAL-001, GVS-STM-001	Wear data for cast polyurethane on quartz, TRL 4	Not verifiable at TRL 3 (STM S29)
UR-011	PR-015	Stability with a full tank	GVS-CAL-001	Analysis, tipping and post check (GVS-CAL-001 section 11), TRL 3	Met on paper: 13.1 degrees with the 560 mm drum (GVS-DEC-001 F1); post safety factor 2.1 at a 185 N push; sway of the full tank near the table drive frequency to decide (GVS-DEC-001 O3)
UR-011	PR-016	Containment of a liner fragment	GVS-CAL-001, GVS-OPT-001	Analysis (GVS-CAL-001 section 10), TRL 3; impact test before any spin test, TRL 4	At risk:** contained face first (57 J against 40 J); not shown edge first (14 J against 22 J at 900 rpm) or for a quarter of the whole bowl group (125 J); band option awaiting Amish (GVS-DEC-001 O2)

Gaps

None.